

Homework 1

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PART 1: Hadoop Installation on macOS (Intel)

1. I started with installing Hadoop using package manager named Homebrew that helped me to make installation process simpler. Then I checked the folder for making sure that the installed Hadoop is in the right place with the right version.

```
brew install hadoop
```

```
hadoop version
```

```
Hadoop 3.4.2
Source code repository https://github.com/apache/hadoop.git -r 84e8b89ee2ebe6923691205b9e171badde7a495c
Compiled by ahmarsu on 2025-08-20T10:30Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum fa94c67d4b4be021b9e9515c9b0f7b6
This command was run using /usr/local/Cellar/hadoop/3.4.2/libexec/share/hadoop/common/hadoop-common-3.4.2.jar
```

```
cd /usr/local/Cellar/hadoop/3.4.2/libexec/etc/hadoop
```

```
ulkarahmadli@hadoop % code .
```

2. After opening Visual Studio Code, I had to make some changes for configuring environment for Hadoop.

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:/usr/local/Cellar/hadoop/3.4.2/libexec/data/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:/usr/local/Cellar/hadoop/3.4.2/libexec/data/datanode</value>
  </property>
</configuration>
```

mapred-site.xml

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

  <property>
    <name>yarn.nodemanager.env-whitelist</name>
    <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PREPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_HOME</value>
  </property>
</configuration>
```

3. I had to install Java 8 for Hadoop, but as AdoptOpenJDK 8 was not available through homebrew, I used Eclipse Temurin 8.

```
brew install --cask temurin@8
```

```
/usr/libexec/java_home -v 1.8
```

```
/Library/Java/JavaVirtualMachines/temurin-8.jdk/Contents/Home
```

4. I opened ~/.zshrc and added the lines below to start HADOOP_HOME for map-reduce jobs.

```
code ~/.zshrc
```

```
Users > ulkarahmadli > $ .zshrc
1  export HADOOP_HOME=/usr/local/Cellar/hadoop/3.4.2/libexec
2  export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
3  export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
4  export YARN_HOME=$HADOOP_HOME
```

5. I started the Hadoop.

```
nijatahmadli@ulkarahmadli hadoop % start-all.sh

WARNING: Attempting to start all Apache Hadoop daemons as nijatahmadli in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ulkarahmadli]
2025-10-14 21:38:40,656 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
resourcemanager is running as process 34189. Stop it first and ensure /tmp/hadoop-nijatahmadli-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 34287. Stop it first and ensure /tmp/hadoop-nijatahmadli-nodemanager.pid file is empty before retry.
```

6. I checked the information about Java processes with “jps” command (Java Virtual Machine Process Status Tool). However, I faced a challenge in that step.

```
37168 Jps
36688 SecondaryNameNode
36548 DataNode
34189 ResourceManager
34287 NodeManager
```

I wasn't able to open <http://localhost:9870> because Hadoop couldn't find access to “NameNode”. So I created and assigned permissions manually, and secured the directories. Then I had to “format” and succeeded.

```
nijatahmadli@ulkarahmadli hadoop % sudo mkdir -p /usr/local/Cellar/hadoop/3.4.2/
libexec/data/namenode
sudo mkdir -p /usr/local/Cellar/hadoop/3.4.2/libexec/data/datanode
sudo chown -R nijatahmadli /usr/local/Cellar/hadoop/3.4.2/libexec/data
sudo chmod -R 755 /usr/local/Cellar/hadoop/3.4.2/libexec/data

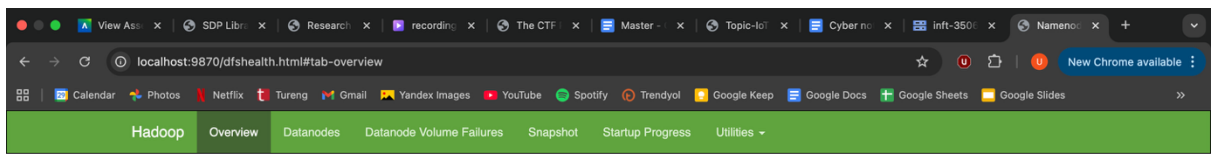
[Password:
nijatahmadli@ulkarahmadli hadoop % hdfs namenode -format

2025-10-14 21:44:41,307 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = ulkarahmadli/127.0.0.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.4.2
STARTUP_MSG:   classpath = /usr/local/Cellar/hadoop/3.4.2/libexec/etc/hadoop:/us
r/local/Cellar/hadoop/3.4.2/libexec/share/hadoop/common/lib/jaxb-impl-2.2.3-1.ja
r:/usr/local/Cellar/hadoop/3.4.2/libexec/share/hadoop/common/lib/curator-client-
5.2.0.jar:/usr/local/Cellar/hadoop/3.4.2/libexec/share/hadoop/common/lib/kerby-p
kix-2.0.3.jar:/usr/local/Cellar/hadoop/3.4.2/libexec/share/hadoop/common/lib/rel
```

Then, I used “jps” command again and got the expected result.

```
nijatahmadli@ulkarahmadli hadoop % jps
38789 Jps
38427 SecondaryNameNode
38619 ResourceManager
38285 DataNode
38157 NameNode
38718 NodeManager
```

After this step, I was able to open <http://localhost:9870> on my browser.



Overview 'localhost:9000' (✓active)

Started:	Tue Oct 14 21:44:59 +0400 2025
Version:	3.4.2, r84e8b89ee2ebe6923691205b9e171badde7a495c
Compiled:	Wed Aug 20 14:30:00 +0400 2025 by ahmarsu from branch-3.4.2
Cluster ID:	CID-a04a6e28-e698-4b29-a6ed-a0b57a984cb7
Block Pool ID:	BP-921571909-127.0.0.1-1760463882039

Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).

Heap Memory used 99.23 MB of 150 MB Heap Memory. Max Heap Memory is 4 GB.

Non Heap Memory used 51.79 MB of 54.13 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	233.47 GB
Configured Remote Capacity:	0 B
DFS Used:	4 KB (0%)
Max DFS Heap:	203.7 GB

At the end, I stopped the Hadoop service.

```
nijatahmadli@ulkarahmadli:~$ hadoop % stop-all.sh

WARNING: Stopping all Apache Hadoop daemons as nijatahmadli in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [ulkarahmadli]
2025-10-14 21:48:38,135 WARN util.NativeCodeLoader: Unable to load native-hadoop
  library for your platform... using builtin-java classes where applicable
Stopping nodemanagers
Stopping resourcemanager
```

PART 2: Hadoop and MapReduce for Wordcount

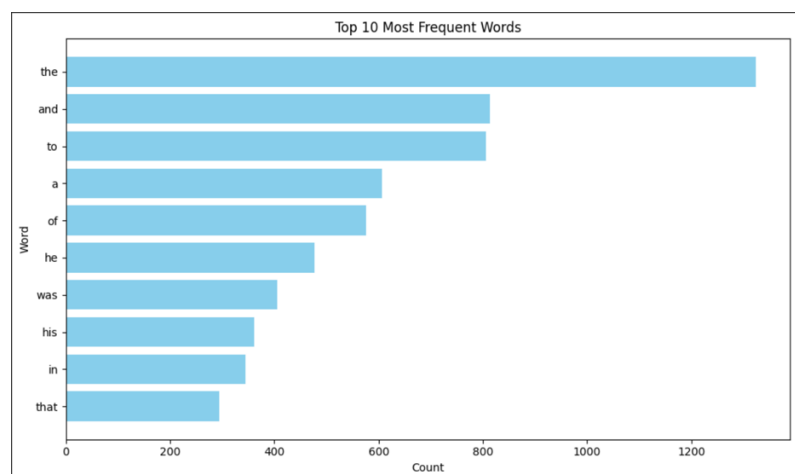
1. I started with downloading a book from <https://www.gutenberg.org/>. I chose the book **“The defender by Nicholas Kalashnikoff”**. I downloaded the plaintext and saved it as a txt file.
2. Then, I moved the book file to the Hadoop system. I did the wordcount with the commands below.

```
nijatahmadli@ulkarahmadli Desktop % hadoop fs -put defender.txt /user/ulkarahmadli/book  
2025-10-14 22:46:44,932 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

```
nijatahmadli@ulkarahmadli Desktop % hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /user/ulkarahmadli/book/defender.txt /user/ulkarahmadli/output
```

3. After copying the output, I prepared Python code and ran it on Google Colab to be able to visualize the results. Here it is:

```
Word Count Summary:  
Total number of words: 27543  
Number of unique words: 5934  
  
🏆 Top 10 Most Frequent Words:  
word count  
the 1323  
and 814  
to 806  
a 607  
of 576  
he 477  
was 405  
his 362  
in 345  
that 295
```



From the word count, I can guess that it is a short novel. Additionally, frequency of pronouns like “he”, “his” gives us the idea that the protagonist is a male. Furthermore, the frequently use of “was” helps us to guess that the story is told in past tense.

PART 3.1: Stopword Removal

1. Firstly, I put `stopword_list` that was provided in instructions to the system.

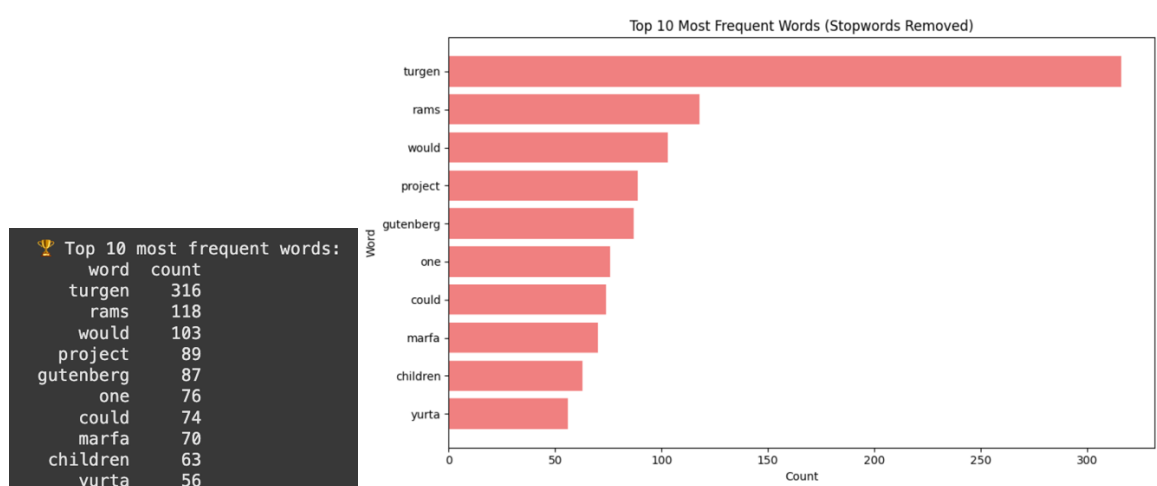
```
hdfs dfs -put ~/Desktop/stopword_list.txt /user/ulkarahmadli/book/
```

2. I used “awk” to remove the stopwords. After that step, I did MapReduce job.

```
nijatahmadli@ulkarahmadli Desktop % awk '
# === Load stopwords ===
BEGIN {
  while (getline < "stopword_list.txt") {
    stopwords[$0] = 1
  }
}

# === Process text ===
{
  for (i = 1; i <= NF; i++) {
    word = tolower($i)           # convert to lowercase
    gsub(/^[^a-zA-Z]/, "", word) # remove punctuation
    if (!(word in stopwords) && word != "")
      printf "%s ", word
    print ""
  }
}' defender.txt > defender_filtered.txt
```

3. Again, I took the output and visualized it. Here are the results:



We can guess that “turgen” is the name of the main character as it dominates the wordcount. Also, some words like “yurta”, “rams” helps us to guess that the story reflects Siberian culture.

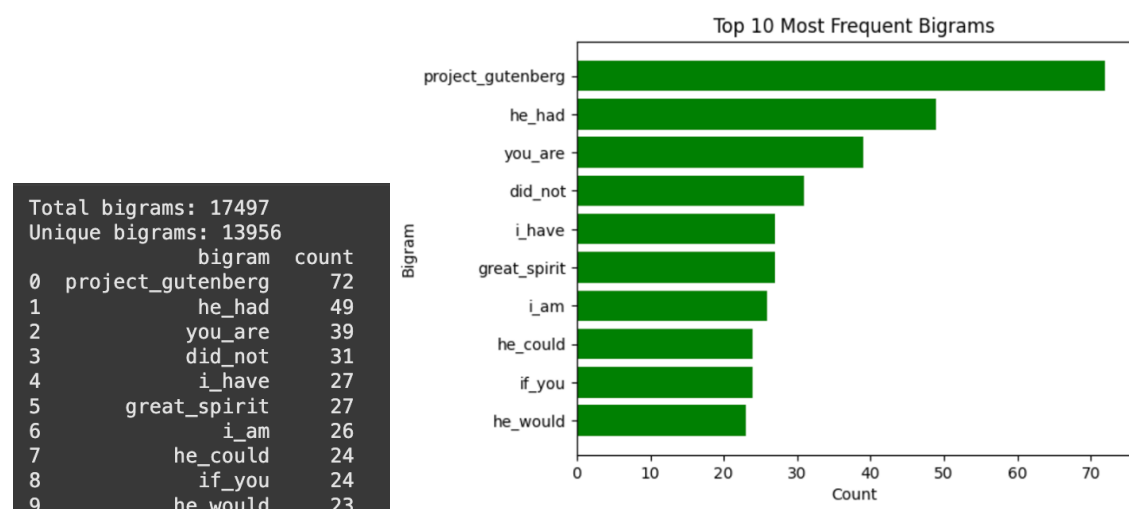
PART 3.2: Bigram Counting

1. As in the stopwords step, I used “awk” for bigram. I entered my code to terminal and got the updated text file.

```
nijatahmadli@ulkarahmadli Desktop % awk '{
  for (i = 1; i < NF; i++) {
    word1 = tolower($i)
    word2 = tolower($(i+1))
    gsub(/^[^a-zA-Z]/, "", word1)
    gsub(/^[^a-zA-Z]/, "", word2)
    if (word1 != "" && word2 != "")
      print word1 "_" word2
  }
}' defender_filtered.txt > defender_bigrams_joined.txt
```

```
nijatahmadli@ulkarahmadli Desktop % hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar \
wordcount /user/ulkarahmadli/bigram_input/defender_bigrams_joined.txt /user/ulkarahmadli/bigram_output
```

2. I used my python code to visualize bigram results. Here it is:



Phrases like “he had”, “he could” again shows as the dominance of male character.

Additionally, we can get the spiritual context of novel with the usage of bigrams like “great_spirit”.